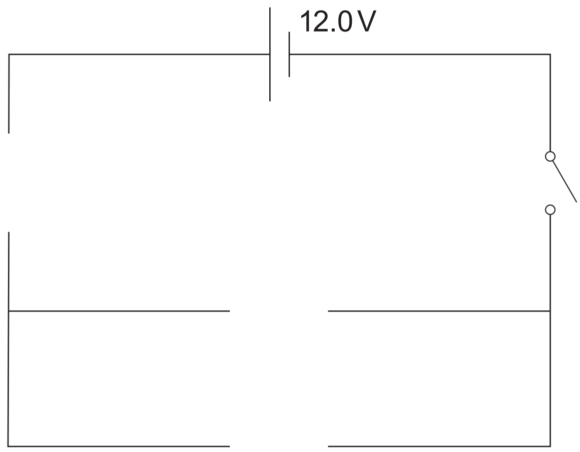
Examiner only

6. A thermistor is a type of resistor whose resistance changes with temperature. Thermistors are used as temperature sensors.

To investigate a thermistor, students set up the following circuit to take measurements of current through the thermistor and the voltage across it.



(a) **Complete** the circuit diagram using the correct circuit symbols. [3]

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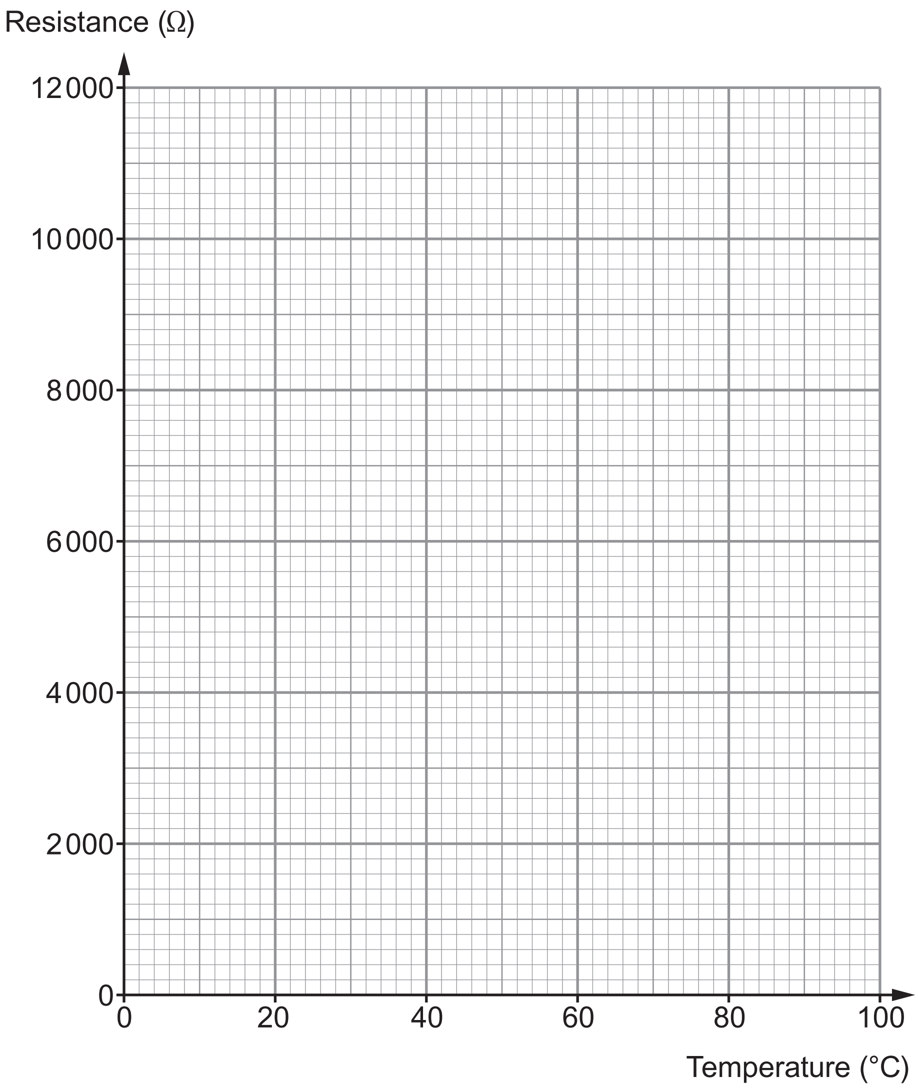
(b) The students place the thermistor in water. They take current and voltage readings at different temperatures. These are used to calculate the resistance of the thermistor at each temperature. The results are shown in the table below.

Temperature (°C)	Current (mA)	Voltage (V)	Resistance (Ω)
20	1.0	12.0	12000
40	2.2	12.0	5400
60	4.8	12.0	2500
80	8.6	12.0	1400
100	20.0	12.0	600

(i) Plot the data on the grid below and draw a suitable line. [3]



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(ii) Describe what happens to the resistance of the thermistor as the temperature increases. [2]

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(c) (i) Use your graph to find the resistance of the thermistor at 50 °C. [1]

Resistance at 50 °C = ... Ω

(ii) Use an equation from page 2 to calculate the current through the thermistor at 50 °C. [3]

Current = ... A

(d) A temperature sensor needs to vary in resistance by at least 3600 Ω as the temperature changes from 40 °C to 80 °C. Use the results from the experiment to explain whether the thermistor used by the students would be suitable. [2]

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